AY 2026-27

AICTE SAMARTHAN INTERNSHIP CONNECT 2026-27

On

How can epistemic change identified in publications with the application of LLMs to full texts of publications

Submitted for
Internship Project

By

AAMIR QADIR SOFI, WASI CHARAK, ANSHIKA SHARMA

Submitted to:-
Dr. Prince Sharma, Faculty
at
SCHOOL OF COMPUTING
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA
HIMACHAL PRADESH, INDIA

Originality / No Plagiarism Declaration

We certify that this internship report entitled **"How can epistemic change be identified in publications through the application of LLMs to full texts of publications "** is our original work. The information presented by us is for academic purposes. Whenever ideas, images, or information from other sources have been used, proper acknowledgment and reference have been provided.

We also certify that this report has not previously been submitted, either in whole or in part, for the award of any other certificate, diploma, or degree.

AAMIR QADIR SOFI
(1VA23CS002)
Sai Vidya Institute Of Technology Rajanukunte Banglore Karnataka

WASI CHARAK
(2419916)
Jamia Hamdard Street, New Delhi Delhi

ANSHIKA SHARMA
(2419916)
Shivaji University Kohlapur, Maharashtra

Bonafide Certificate

This is to certify that the project titled **"How can epistemic change be identified in publications through the application of LLMs to full texts of publications "** submitted by **AAMIR QADIR SOFI (1VA23CS002), WASI CHARAK (2419916), ANSHIKA SHARMA (2419916)** is a bonafide record of the work carried out during the internship in partial fulfilment of the requirements of the internship program at **Indian Institute of Information and Technology Una, Himachal Pradesh, India.**

AAMIR QADIR SOFI (1VA23CS002)	**WASI CHARAK (2419916)**	**ANSHIKA SHARMA (2419916)**
Sai Vidya Institute Of Technology Rajanukunte Banglore Karnataka	Jamia Hamdard Street, New Delhi Delhi	Shivaji University Kohlapur, Maharashtra

Supervised by:
Dr. Prince Sharma
School of Computing
Indian Institute of Information Technology Una
Himachal Pradesh, India

Contents

Originality / No Plagiarism Declaration 1

Bonafide Certificate 2

Acknowledgement 5

1 Introduction 7

2 Literature Survey 9
2.1 Large Language Models for Scientific Text Analysis 10
2.2 Epistemic Change and Knowledge Evolution 10
2.3 Bibliometric and Full-Text Mining Methods 11

3 Methods 12
3.1 Research methodology 13
3.2 Study Selection 15

4 Results 16

5 Conclusion 19

6 Carried Out Work 20
6.1 Sankey Diagram Using R 21
6.2 Report Preparation Using LATEX on Overleaf 21
6.3 Excel Dataset for Keyword Screening 22
6.4 GitHub Repository 23

Abstract

The identification of epistemic change within scientific publications remains a critical yet underexplored challenge in bibliometric research. This study examines how Large Language Models (LLMs) can be applied to the full texts of scientific publications to detect epistemic change, knowledge evolution, and conceptual shifts across research domains. A systematic literature search, guided by the PRISMA 2020 framework, was conducted across eight major scholarly databases — Web of Science, Google Scholar, SpringerLink, Wiley Online Library, IEEE Xplore, ACM Digital Library, ScienceDirect, and Scopus —using a Boolean query constructed from four core semantic attributes: Large Language Models, epistemic change, full-text analysis, and scientific publications. The query was formulated using principles from automata theory, combining synonymous terms through logical AND/OR operators to maximize retrieval coverage while preserving semantic precision. Retrieved records were screened, de-duplicated, and organized into a structured dataset. Each paper was first classified by LLMs (ChatGPT and Claude) for the presence of the four target constructs, and the same papers were subsequently screened manually at two levels: an abstract-level pass and a full-text-level pass. Comparing these three screening layers revealed measurable discrepancies in construct identification between AI-generated, abstract-only, and full-text-level classifications, indicating that LLMs and human reviewers detect epistemic-change indicators differently depending on the depth of text examined. These findings underscore the limitations of abstract-only bibliometric screening and demonstrate the added value of full-text analysis for capturing genuine epistemic change. The study contributes a reproducible, LLM-assisted workflow for tracing knowledge evolution in scientific literature and offers practical guidance for future bibliometric and scientometric research combining AI-driven and traditional screening methods.

Keywords

Large Language Models (LLMs), Epistemic Change, Full-Text or Text Mining, Publications

Acknowledgement

We would like to express our sincere gratitude to the **All India Council for Technical Education** (AICTE) for providing us with the opportunity to undertake this internship through the **AICTE SAMARTHAN INTERNSHIP CONNECT 2026-2027** Initiative, which enhanced our practical knowledge and understanding in the field of bibliometric research, large language models, and scientific literature analysis.

We express our heartfelt thanks to our Nodal Officer, Dr. Ravneet Singh, for her constant support, encouragement, and valuable guidance throughout the internship.

We are grateful to our Mentor, Dr. Prince Sharma, School of Computing, Indian Institute of Information Technology (IIIT) Una, Himachal Pradesh, for his valuable guidance, continuous encouragement, and insightful suggestions during the internship and while preparing this report. His support greatly contributed to the successful completion of this internship and this report.

We would also like to thank IIIT Una for providing a conducive learning environment and the necessary resources to successfully complete this internship. Finally, we express our gratitude to everyone who directly or indirectly contributed to making this internship a valuable learning experience.

AAMIR QADIR SOFI
(1VA23CS002)
Sai Vidya Institute Of Technology Rajanukunte Banglore Karnataka

WASI CHARAK
(2419916)
Jamia Hamdard Street, New Delhi Delhi

ANSHIKA SHARMA
(2419916)
Shivaji University Kohlapur, Maharashtra

List of Figures

1 Distribution of reviewed publications by year. 11
2 Distribution of reviewed publications by publisher. 12
3 Review process workflow. 13
4 PRISMA 2020 flow diagram showing the literature selection process. 16
5 Results of AI based interpretation. 17
6 Distribution of Correct and Incorrect Inference Results 19
7 Sankey diagram showing the flow of screened papers from source databases through LLM, Epistemic Change, and Full-text/Text Mining screening criteria. 21
8 Screenshot of the Overleaf LaTeX editor used for preparing the internship report using the Springer Nature `sn-jnl` template. 22
9 Screenshot of the Excel dataset used for recording keyword occurrences across three screening levels: AI-generated, manual screening till abstract, and manual screening after abstract. . . 23
10 Screenshot of the GitHub repository 24

CHAPTER 1: Introduction

Scientific knowledge continuously evolves as new discoveries, methodologies, and technologies reshape existing understanding across disciplines. This process, commonly referred to as *epistemic change*, reflects the development, refinement, replacement, or expansion of knowledge over time. Identifying epistemic change within scientific publications has become increasingly important for researchers, policymakers, funding agencies, and academic institutions because it enables them to understand how research fields emerge, mature, and transform. Traditionally, the identification of such changes has relied on citation analysis, manual literature reviews, and expert evaluation. Although these approaches provide valuable insights, they are often time-consuming, difficult to scale, and limited in their ability to capture subtle semantic changes occurring within the full text of scientific publications.

Recent advancements in Artificial Intelligence (AI), particularly Large Language Models (LLMs), have opened new possibilities for analyzing scientific literature at an unprecedented scale. LLMs such as GPT, LLaMA, and other transformer-based models possess advanced natural language understanding capabilities that enable them to process, summarize, compare, and interpret large volumes of textual information. Unlike conventional bibliometric approaches that primarily focus on metadata such as citation counts, publication years, author collaborations, and keywords, LLMs are capable of analyzing the complete textual content of research articles. Consequently, they can identify semantic evolution, conceptual shifts, emerging research themes, and changes in scientific terminology that may not be evident through traditional bibliometric indicators alone.

Bibliometric analysis has become one of the most widely adopted quantitative techniques for evaluating scientific research performance and understanding knowledge structures within academic disciplines. It employs statistical methods to analyze publication metadata, citation relationships, co-authorship patterns, institutional collaborations, keyword co-occurrence, and publication trends. Combined with systematic literature review methodologies such as the PRISMA 2020 framework, bibliometric analysis provides a transparent and reproducible process for identifying relevant scientific publications while minimizing selection bias. The integration of bibliometric techniques with Artificial Intelligence and LLM-based full-text analysis represents a promising direction for understanding how scientific knowledge evolves across different domains.

The primary objective of this study is to investigate how Large Language Models can contribute to the identification of epistemic change through the bibliometric analysis and full-text examination of scientific publications. A systematic literature search was conducted across multiple scholarly databases using carefully designed Boolean search queries incorporating keywords related to epistemic change, knowledge evolution, bibliometrics, scientometrics, and Large Language Models. During this internship, SpringerLink and Wiley Online Library were assigned as the primary databases for data collection, and relevant publications were screened according to predefined inclusion and exclusion criteria following the PRISMA 2020 guidelines. Metadata extracted

from the selected publications were organized for subsequent bibliometric analysis.

The findings of this research are expected to provide insights into the emerging role of LLMs in scientometric research and demonstrate how AI-assisted full-text analysis can complement traditional bibliometric techniques for identifying epistemic change. Furthermore, the study aims to establish a reproducible workflow that integrates systematic literature review methods, metadata analysis, and Artificial Intelligence for future research in knowledge evolution studies. The outcomes may benefit researchers, librarians, policy-makers, and academic institutions seeking more efficient approaches to analyze scientific publications and monitor the evolution of knowledge across research domains.

The rapid rise of Large Language Models (LLMs) has opened new possibilities for studying how scientific knowledge evolves, yet their application to tracing epistemic change within scholarly literature remains largely unexplored. Epistemic change — the way knowledge claims are formed, revised, and reshaped within a field — represents a critical lens through which we can examine not just what scientists discover, but how their conceptual frameworks and understanding of evidence transform over time. Traditionally, scholars have relied on manual content analysis, citation patterns, and surface-level bibliometric indicators such as citation counts and keyword frequency to track such evolution. However, these conventional methods suffer from significant limitations: they are labor-intensive, fail to capture nuanced shifts in argumentation or conceptual framing, and struggle to process the exponentially growing volume of scientific literature. Moreover, traditional approaches tend to focus on explicit metadata rather than the rich semantic and contextual information embedded within the full texts of publications themselves. LLMs are capable of processing and interpreting large volumes of unstructured text with increasing sophistication, offering a powerful tool for identifying epistemic shifts with greater depth and precision than earlier approaches. The ability to analyze complete publication texts — rather than abstracts or metadata alone — enables detection of subtle but significant changes in how knowledge is constructed and validated across scientific fields. Despite growing interest in LLM applications across research domains, few studies have specifically examined their potential in analyzing full-text publications to detect epistemic change in a systematic and rigorous manner. This represents a clear gap in the literature that the present study aims to address, by exploring how computational text analysis can illuminate the evolution of scientific knowledge.

The primary objective of this study is to investigate how epistemic change can be systematically identified in scientific publications through the application of LLMs to full-text analysis. To achieve this, the research began with a thorough analysis of the study title into four core keywords — epistemic change, LLM, full-text, and publication. Rather than relying solely on these primary terms, synonyms and alternative expressions were identified for each keyword to maximize search coverage and capture the broadest range of relevant literature. This led to the construction of a comprehensive Boolean search query combining these terms: ("epistemic change" OR "knowl-

edge change" OR "knowledge evolution") AND ("large language model" OR "LLM" OR "LLMs") AND ("full text" OR "full-text analysis" OR "text mining") AND ("publication" OR "publications" OR "scientific literature"). This strategically designed query was then applied consistently across eight major academic databases, including Google Scholar, Wiley, Web of Science, IEEE Xplore, ScienceDirect, ACM Digital Library, and Scopus. The multi-database approach was essential for capturing the widest possible representation of relevant literature, though the search process did encounter various complications during implementation. Despite these challenges, the search strategy successfully retrieved a substantial pool of records for further processing.

Following data retrieval, a rigorous multi-stage screening and preparation process was undertaken to refine the dataset according to predefined inclusion and exclusion criteria. First, papers were filtered based on downloadability and accessibility, as full-text access was essential for meaningful analysis. Second, document type and language restrictions were applied: only English-language journal articles were retained, while conference papers, books, and proceedings were excluded to maintain methodological consistency and focus on peer-reviewed scholarly publications These initial filters resulted in 313 records that met the baseline inclusion criteria. Subsequently, all 313 papers were systematically analyzed against the four core keywords to determine relevance to the study's core objective. To ensure robustness and validity, this relevance assessment was conducted using two complementary approaches: AI-assisted screening methods were first applied, followed by independent manual review. The use of both methods revealed varying results between automated and human analysis, which highlighted important limitations in relying solely on a single analytical approach and provided valuable insight into the strengths and constraints of each method, ultimately allowing for more rigorous cross-validation. Notably, no temporal restrictions were applied during the search or screening process, allowing the final dataset to include publications across all available years in the databases, thus capturing the full historical evolution of research in this area. This comprehensive filtering and screening process successfully narrowed the 313 records down to a final dataset of 191 records deemed most suitable for in-depth analysis. The refined dataset was subsequently processed using Biblioshiny to generate bibliometric visualizations and analyze patterns within the literature, while a PRISMA flow diagram was constructed to document and transparently report the entire selection process. This paper presents the findings derived from this structured methodology and reflects on what they reveal about the current state of research applying LLMs to the study of epistemic change in scientific publications.

CHAPTER 2: Literature Survey

This section reviews existing studies related to the three core aspects of this research: the use of Large Language Models (LLMs) for scientific text analysis, bibliometric and scientometric approaches to knowledge evolution, and full-text mining of scientific publications.

2.1 Large Language Models for Scientific Text Analysis

[11] explore the research landscape on single-case design methodology using text mining and Large Language Models. The authors apply LLMs together with bibliometric analysis to investigate how research topics evolve over time, making this study one of the most closely aligned with the objectives of the present review.[15] present a comprehensive survey of more than 260 scientific Large Language Models developed across different scientific fields. Although the paper explains in detail how these LLMs are trained and applied, it does not investigate how scientific knowledge changes over time and therefore falls outside the strict inclusion criteria of this review.[5] propose a framework for structured information extraction from scientific text using GPT and Llama. While the approach demonstrates strong capabilities in extracting entities and relationships from full-text publications, the study does not address epistemic change or knowledge evolution.[3] introduce SciBERT, a pretrained language model specifically designed for scientific text. The model achieves strong performance on several scientific NLP benchmarks; however, the work focuses on improving language representations rather than identifying changes in scientific knowledge.[12] propose a continuous knowledge metabolism framework that uses LLMs to process full-text scientific papers and explicitly model knowledge evolution over time. The system generates new scientific hypotheses from evolving literature, making it highly relevant to the objectives of this study.[1] develop a dynamic knowledge graph approach integrating temporal embeddings and LLMs to forecast the evolution of scientific knowledge. The system includes full-text mining of publications and explicitly forecasts how scientific knowledge changes, which closely matches the research objective of the present study.

2.2 Epistemic Change and Knowledge Evolution

[6] use text analysis to quantify the similarity and evolution of scientific disciplines. The paper directly studies how scientific disciplines evolve using text mining techniques and is highly relevant to the concept of epistemic change; however, the study does not employ Large Language Models.[9] predict the future impact and evolution of scientific concepts using full-text features extracted from millions of scientific papers. The work studies scientific concept evolution, which is closely related to knowledge evolution, but does not use LLMs for this purpose.[7] apply Latent Dirichlet Allocation (LDA) topic modeling to analyze how research topics in the field of Knowledge Organization change over time. The paper studies knowledge evolution through topic analysis but does not employ transformer-based or Large Language Models.[8] de-

velop computational methods for modeling knowledge evolution and linking hypotheses to evidence from the full text of scientific publications. The study also evaluates modern NLP and LLM-based methods for scientific knowledge synthesis, making it one of the most relevant works identified in this review.

2.3 Bibliometric and Full-Text Mining Methods

[13] propose an integrated LDA and SciBERT model for topic analysis of network science publications. The paper uses SciBERT and text mining for topic discovery but does not explicitly study epistemic change or knowledge evolution, placing it at the boundary of inclusion criteria for this review.[14] identify and visualize key topics in scientific publications using transformer-based language models and document clustering methods. The paper uses GPT-3 to discover research topics and emerging themes, which is close to studying how scientific knowledge evolves over time.[2] introduce the SciQA scientific question answering benchmark for scholarly knowledge. The paper studies scholarly knowledge, mentions knowledge evolution, and evaluates ChatGPT on scientific tasks, making it closely related to the objectives of this review, though it does not directly identify epistemic change.[4] propose a scientific document summarization framework using citation context and scientific discourse. The paper discusses epistemic value drift in citations and analyzes full scientific papers, but its primary goal is summarization rather than the identification of epistemic change using LLMs.[10] develop facet embeddings for explorative analytics in digital libraries by extracting research facets from full-text scientific papers. The work visualizes how research topics and trends change over time and studies research evolution, but does not employ LLMs for this purpose.

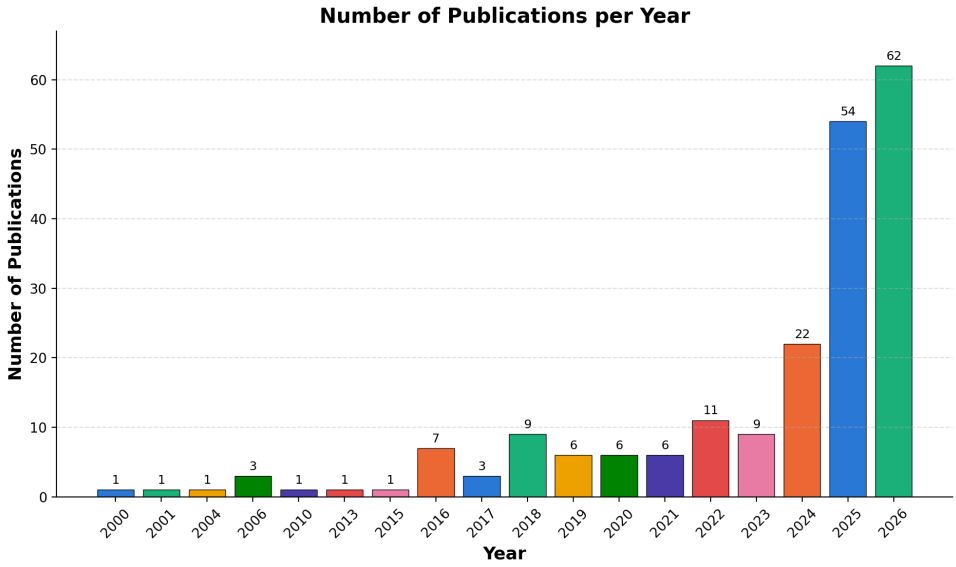


Figure 1: Distribution of reviewed publications by year.

Figure 1 presents the distribution of the reviewed publications by year, illustrating the temporal concentration of research relevant to LLM-based scientific text analysis and knowledge evolution. The chart shows a marked

increase in relevant publications from 2022 onward, with the number of studies rising sharply in 2024, 2025, and 2026. This trend reflects the rapid growth of interest in applying Large Language Models to scientific literature analysis following the widespread adoption of transformer-based architectures and generative AI tools. The scarcity of publications before 2016 further indicates that this research area is comparatively young, with most foundational and applied work emerging only in the past five years.

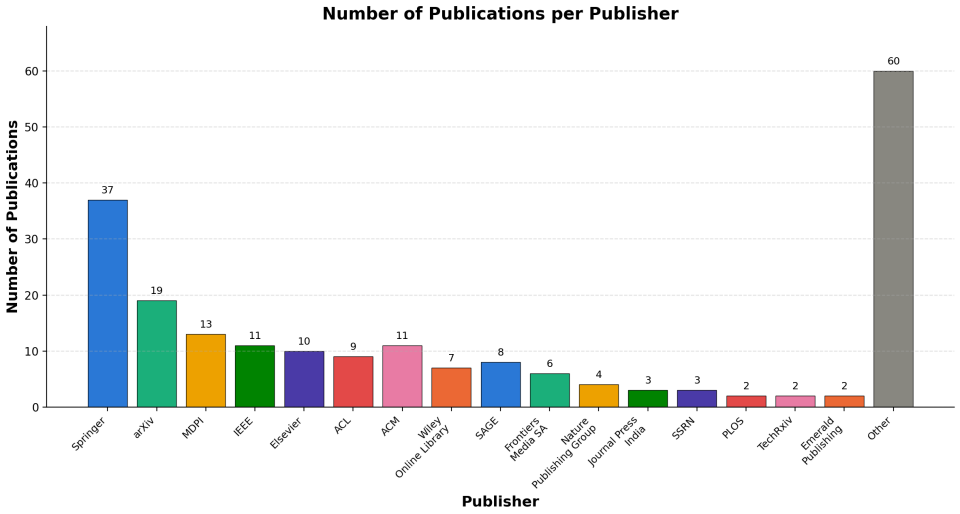


Figure 2: Distribution of reviewed publications by publisher.

Figure 2 shows the distribution of the reviewed publications across major publishers. Springer accounts for the largest share of the reviewed literature, followed by preprint repositories such as arXiv, reflecting the fast-moving nature of research on LLMs and scientific knowledge evolution, where results are often shared ahead of formal peer review. MDPI, IEEE, and Elsevier also contribute a substantial portion of the reviewed studies, indicating broad interest in this topic across open-access and traditional engineering-focused venues. The "Other" category aggregates publishers that appear only once in the reviewed set, reflecting the interdisciplinary and fragmented nature of this emerging research area.

CHAPTER 3: Methods

This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines to ensure a transparent, systematic, and reproducible literature selection process. The methodology consisted of four major phases: identification, screening, eligibility assessment, and final inclusion of relevant studies.

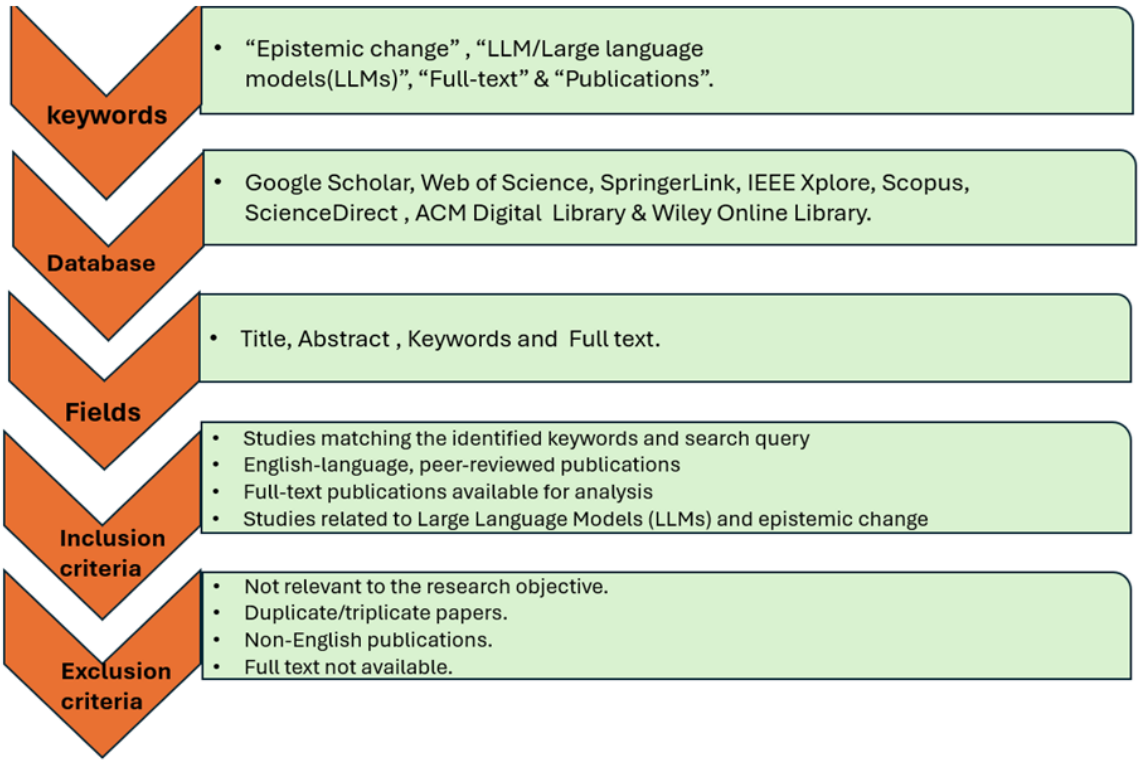


Figure 3: Review process workflow.

3.1 Research methodology

A systematic screening methodology was adopted to identify and select relevant research papers for the final analysis of this study, which investigates the application of Large Language Models (LLMs) to identify epistemic change from full-text publications. The screening process followed a structured, multi-stage approach to ensure that only relevant, accessible, and methodologically appropriate studies were included . Initially, a comprehensive search strategy was developed using a predefined Boolean search query based on the research objectives and key concepts. The search was conducted across multiple academic databases. During the retrieval process, differences in the search functionalities of individual databases were observed. While some databases supported the complete Boolean search query, others required simplified keyword-based searches or modifications in the use of Boolean operators (AND/OR), quotation marks, wildcards, and search syntax. Therefore, minor adjustments were made to the query format for each database while preserving the original search intent, eligibility criteria, and scope of the study to maximize the retrieval of relevant publications . Following the search, all retrieved records

were exported and screened according to predefined inclusion and exclusion criteria. In the first stage of screening, studies were excluded if the full text was inaccessible because of paywalls or restricted access. Papers were also excluded if they were irrelevant based on title screening or if their abstracts did not address the research objectives. Studies with incomplete bibliographic information or unavailable abstracts were removed because they could not be evaluated reliably. Furthermore, studies involving Large Language Models (LLMs), BERT, BERTopic , or other Natural Language Processing (NLP) models were excluded when they did not analyse full-text publications, as full-text analysis was a mandatory requirement of this research. Publication types such as books, conference proceedings, editorials, surveys, datasets, author indexes, magazine articles, reports, and other non-peer-reviewed or non-research documents were also excluded to ensure consistency and methodological quality . After the initial screening, duplicate and triplicate records retrieved from multiple databases were identified and removed to eliminate redundancy. The remaining studies were then screened further, and non-English publications were excluded to maintain consistency in language and interpretation. A final abstract screening was conducted to verify that each study directly addressed the research objectives, aligned with the inclusion criteria, and demonstrated the use of appropriate methods relevant to the study. Only studies that satisfied all predefined eligibility criteria were retained for the final dataset . This systematic, transparent, and multi-stage screening process ensured that the final set of publications was relevant, accessible, original, and methodologically rigorous. The application of predefined screening criteria at each stage minimized selection bias, improved the reliability and reproducibility of the review process, and provided a strong foundation for the subsequent analysis of the selected studies.

A comprehensive literature search was conducted across eight major scholarly databases, namely Springer, Wiley Online Library, ACM Digital Library, IEEE Xplore, Scopus, Web of Science, ScienceDirect, and Google Scholar The following Boolean search query was applied consistently across all databases:

("epistemic change" OR "knowledge change" OR "knowledge evolution")
AND
("large language model" OR "LLM" OR "LLMs")
AND
("full text" OR "full-text analysis" OR "text mining")
AND
("publication" OR "publications" OR "scientific literature")

The search was limited to English-language publications. Metadata including title, authors, publication year, DOI, journal, URL, and publication type were extracted for each retrieved study.The Boolean query was designed as a structured expression in which each group of terms represents a distinct semantic attribute of the research problem. The first group captures epistemic change through related concepts such as knowledge change and knowledge evolution. The second group identifies studies involving Large Language Models, while the third group focuses on full-text analysis and text-mining approaches.

The final group restricts the search to scientific publications and related literature. From an automata-theory perspective, the query can be viewed as a formal expression composed of valid terms connected through Boolean operators. The `OR` operator expands the search by accepting alternative terms within the same conceptual attribute, whereas the `AND` operator combines multiple attributes and retrieves only those records that satisfy all required conditions. This structured formulation improves semantic relevance, reduces ambiguity, and supports the systematic retrieval of publications relevant to the study.

3.2 Study Selection

After collecting the search results, duplicate records were removed and the remaining papers were screened based on their titles and abstracts. Papers that appeared relevant to the research objective were then assessed through full-text reading. The following inclusion criteria were adopted:

- Studies involving Large Language Models (LLMs) or transformer-based language models.
- Studies related to epistemic change, knowledge evolution, scientific concept evolution, or closely related topics.
- Studies using full-text analysis, text mining, or bibliometric analysis of scientific publications.
- Peer-reviewed journal articles and conference papers written in English.

The exclusion criteria included:

- Papers focusing only on document summarization, recommendation systems, or information retrieval without knowledge evolution analysis.
- Studies unrelated to scientific publications.
- Editorials, short communications, tutorials, or duplicate records.

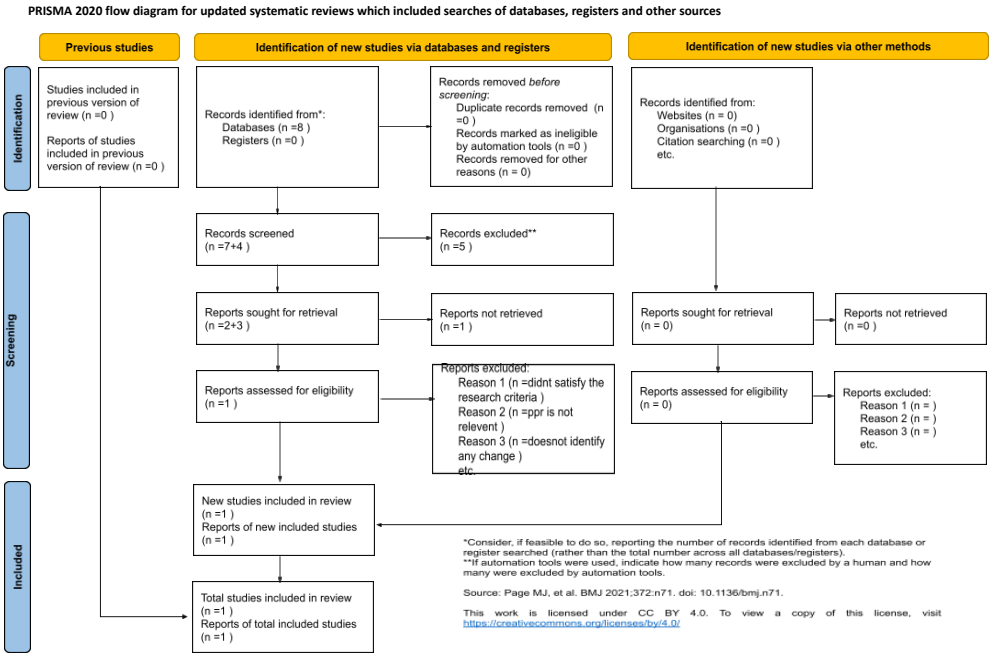


Figure 4: PRISMA 2020 flow diagram showing the literature selection process.

CHAPTER 4: Results

Completing the above anlaysis we are able derive successful derivation in the use of LLMs for identifying epistemic changes. Based on the provided research remarks, we are able to derive the following list of conclusions that can be derived regarding how epistemic change is identified using Large Language Models (LLMs) on the full text of publications:

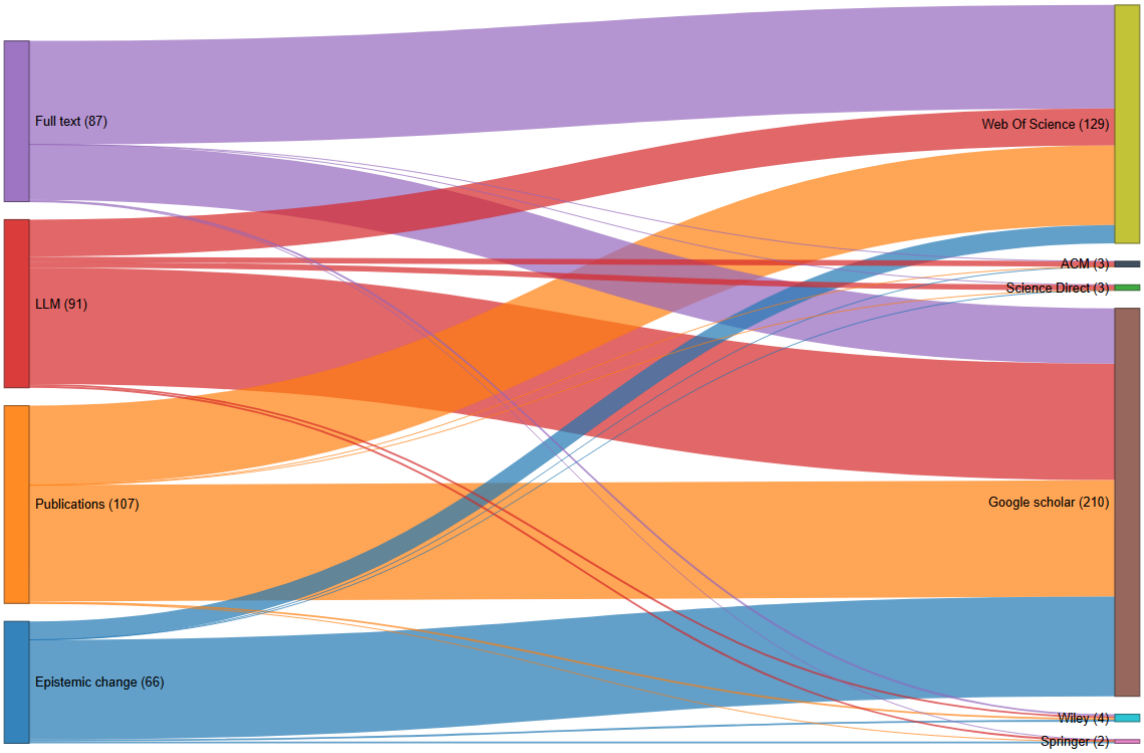


Figure 5: Results of AI based interpretation.

Fig 5 The Sankey diagram presents the AI-based interpretation of the retrieved publications by illustrating the relationship between the selected keywords and the databases from which the publications were obtained. The keywords Publications, LLM, Full Text, and Epistemic Change are connected to databases such as Google Scholar, Web of Science, ScienceDirect, ACM Digital Library, IEEE Xplore, and Springer. The width of each flow represents the number of publications associated with each keyword–database combination. The diagram indicates that Google Scholar and Web of Science contributed the largest share of relevant publications, while the remaining databases contributed comparatively fewer records. This visualization provides a clear overview of the distribution of literature identified through the AI-based interpretation process.

1. **Integration of LLMs with Bibliometrics and Text Mining.**
   To successfully capture how scientific knowledge and research topics evolve over time, effective frameworks combine the semantic capabilities of LLMs (such as BERT, SciBERT, or generative models) with tra-

ditional text mining and bibliometric analysis. This hybrid approach allows researchers to map out broad structural shifts alongside deep contextual changes.

2. **Necessity of Full-Text Analysis Over Metadata**
   A recurring theme in the literature is that analyzing titles, abstracts, and metadata (e.g., using standard LDA or basic BERTopic) is insufficient for capturing true epistemic change. Explicitly identifying complex knowledge shifts and paradigm variations requires deep full-text mining to evaluate the complete discourse, context, and argumentation of scientific writing.
3. **Dynamic Knowledge Modeling and Hypothesis Generation**
   When applied to full-text publication streams, LLMs can explicitly model knowledge evolution dynamically over time. By analyzing sequentially published literature, these systems can track how concepts transform, allowing them to automatically extract underlying evidence and generate new scientific hypotheses.
4. **Detection of Paradigm Shifts and Emerging Themes**
   Advanced language models excel at semantic representation, which enables them to look beyond simple keyword frequencies. They are used to analyze topic evolution, identify sudden paradigm shifts, track conceptual changes, and forecast the future trajectory of scientific knowledge within specific fields.
5. **Scientific Knowledge Synthesis and Evidence Extraction**
   Using LLMs on full-text articles allows for advanced scientific knowledge synthesis. This computational modeling helps extract the relationships between hypotheses and evidence, tracing how ideas are recombined, verified, or altered across a vast body of literature over time.

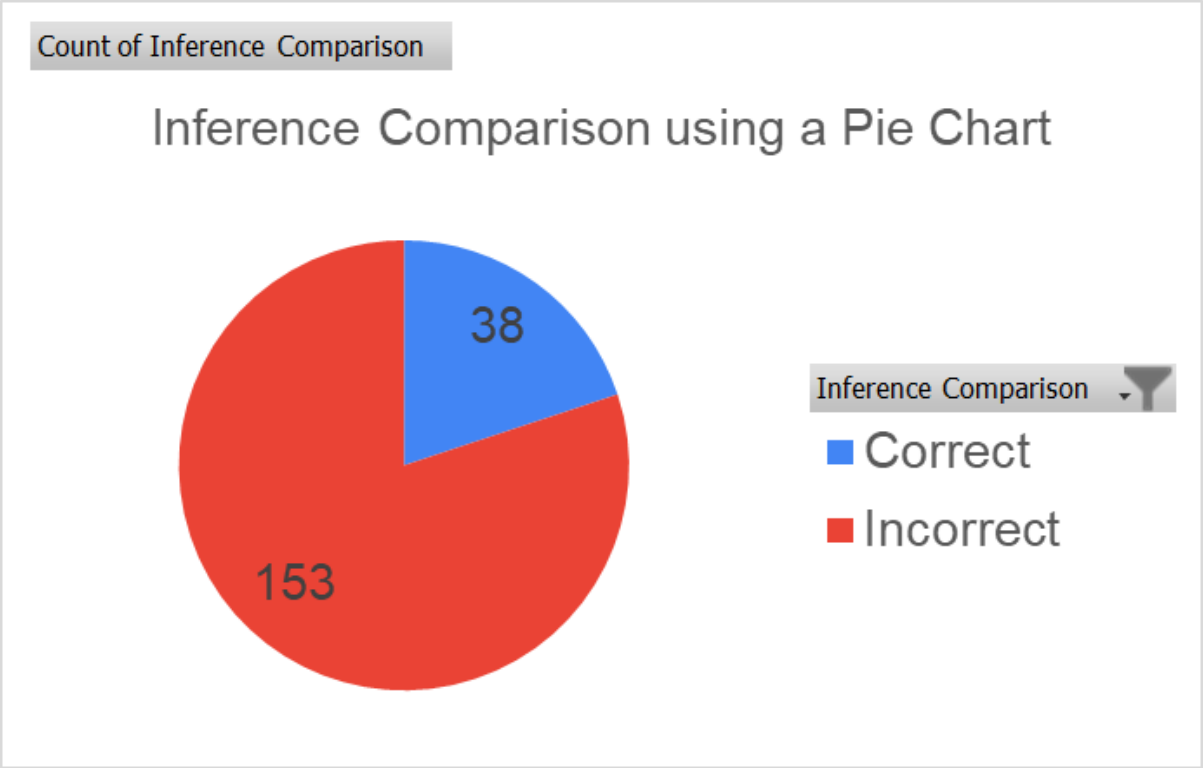


Figure 6: Distribution of Correct and Incorrect Inference Results

Fig 6 Illustrates the inference comparison evaluates the performance of the AI-based interpretation by comparing the generated inference results with the manually verified classifications. The pie chart shows that 38 inferences were correct, while 153 were incorrect. This indicates that the AI model was able to correctly identify only a limited number of cases, whereas most of the inferences did not match the manual evaluation. The comparison highlights the importance of validating AI-generated interpretations with human expertise and suggests that further refinement of the inference process is required to improve its accuracy and reliability.

CHAPTER 5: Conclusion

This systematic review reveals that research combining LLMs with epistemic change analysis is still in its nascent stages, indicating significant untapped potential for this methodological approach. The refinement of 313 records to 191 through rigorous screening demonstrates both the feasibility and the challenges of applying computational methods to bibliometric analysis. The varying results between AI-assisted and manual review underscore a critical finding: while LLMs can efficiently process large volumes of full-text data, their application in detecting epistemic nuance requires careful human oversight and validation. This study establishes that a hybrid approach—combining automated text analysis with human expertise—is essential for reliable identification of knowledge evolution in scientific literature. However, this research encountered several significant limitations that constrained the scope and comprehensiveness of the final dataset. The initial pool of 313 records had to be further refined due to large data volume requiring multiple rounds of deduplication to remove duplicate entries across different databases, resulting in the exclusion of numerous redundant records. Language restrictions eliminated all non-English publications, introducing a potential language bias that may under represent research from non-English speaking regions and limiting the global perspective of the findings. Additionally, the decision to include only journal articles necessitated the exclusion of conference papers, books, and book chapters—formats that often contain significant scholarly contributions and emerging ideas. Paywall restrictions and accessibility constraints further limited the dataset, as papers that could not be downloaded in full text were necessarily excluded, potentially biasing the sample toward open-access or widely-distributed publications. These cumulative filtering stages, while necessary for methodological consistency, mean the final 191 records represent a narrowed view of the complete landscape of research on epistemic change and LLMs. Future studies must address these limitations by implementing broader language coverage, incorporating conference proceedings and grey literature, and finding solutions to access barriers to capture a more comprehensive and globally representative picture of epistemic change research.

CHAPTER 6: Carried Out Work

This section presents the key tools, technologies, and datasets developed and used during the internship to address the research objective of identifying epistemic change in scientific publications using Large Language Models.

6.1 Sankey Diagram Using R

To visualize the flow of retrieved papers through the screening stages, a Sankey diagram was developed using the R programming language. The `networkD3` package was used to construct the diagram, which illustrates how papers from each source database passed through the three screening criteria, namely LLM, Epistemic Change, and Full-text/Text Mining. The `dplyr` package was used for data manipulation and `htmlwidgets` was used to export the diagram as a standalone interactive HTML file. The final diagram was subsequently exported as a PNG image for inclusion in this report. Figure 7 shows the resulting Sankey diagram depicting the screening flow across all databases.

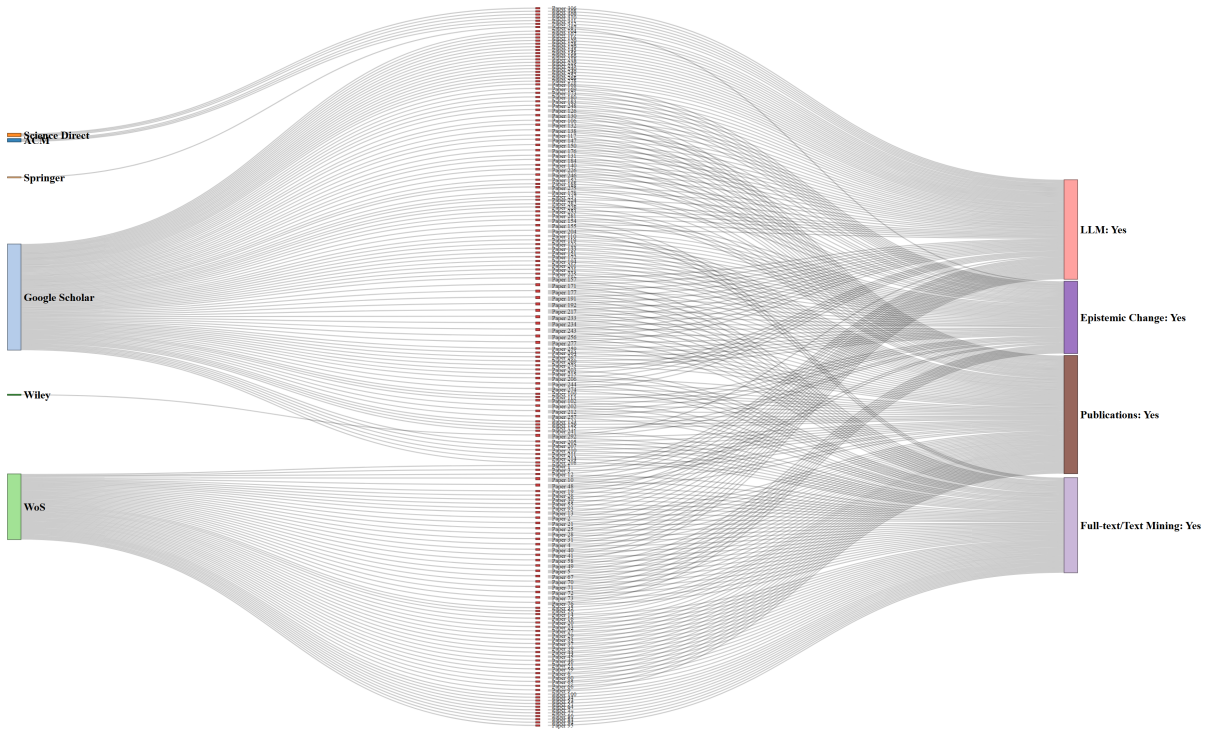
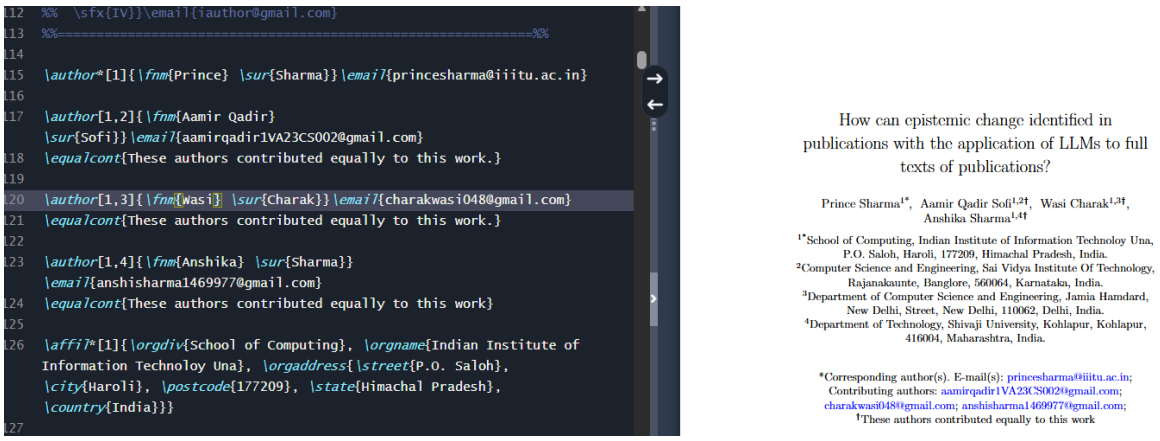


Figure 7: Sankey diagram showing the flow of screened papers from source databases through LLM, Epistemic Change, and Full-text/Text Mining screening criteria.

6.2 Report Preparation Using LATEX on Overleaf

The internship report was prepared using LATEX, a professional document preparation system widely used in academic and scientific publishing. The report was compiled on the Overleaf platform, which provides a cloud-based LATEX editor with real-time compilation and collaborative editing features. The Springer Nature journal template (`sn-jnl`) was used as the document

class to ensure the report conforms to the formatting requirements of Springer publications. All sections, tables, figures, references, and the PRISMA 2020 flow diagram were typeset directly in LATEX code. Figure 8 shows a screenshot of the Overleaf workspace used during the preparation of this report.



How can epistemic change identified in publications with the application of LLMs to full texts of publications?

Prince Sharma[1*], Aamir Qadir Sofi[1,2†], Wasi Charak[1,3†], Anshika Sharma[1,4†]

[1*]School of Computing, Indian Institute of Information Technoloy Una, P.O. Saloh, Haroli, 177209, Himachal Pradesh, India.
[2]Computer Science and Engineering, Sai Vidya Institute Of Technology, Rajanakaunte, Banglore, 560064, Karnataka, India.
[3]Department of Computer Science and Engineering, Jamia Hamdard, New Delhi, Street, New Delhi, 110062, Delhi, India.
[4]Department of Technology, Shivaji University, Kohlapur, Kohlapur, 416004, Maharashtra, India.

*Corresponding author(s). E-mail(s): princesharma@iiitu.ac.in; Contributing authors: aamirqadir1VA23CS002@gmail.com; charakwasi048@gmail.com; anshisharma1469977@gmail.com;
†These authors contributed equally to this work

Figure 8: Screenshot of the Overleaf LATEX editor used for preparing the internship report using the Springer Nature `sn-jnl` template.

6.3 Excel Dataset for Keyword Screening

A structured Excel dataset was developed to record and compare keyword occurrences across all retrieved papers. The dataset consisted of three separate files, each recording the presence or absence of the four core keywords, namely Large Language Models, Epistemic Change, Full-text/Text Mining, and Publications, using a binary classification scheme (1 = present, 0 = absent). The first file was generated by feeding the papers to Large Language Models including ChatGPT and Claude, which automatically detected keyword presence across the full document. The second file was produced through manual screening by the research team up to and including the abstract section of each paper. The third file recorded keyword occurrences found beyond the abstract, covering the introduction, methods, results, and discussion sections. A comparative analysis of all three files revealed measurable discrepancies in keyword counts, providing empirical evidence of how LLMs and human reviewers identify epistemic change indicators differently depending on the depth of text analyzed. Figure 9 shows a screenshot of the Excel dataset used for keyword screening.

A	B	C	D	E	F	G	H	I
Sl.NO	Name	Article Title	LLM	Epistemic Chan	Full-text/Text M	Downloaded	Remarks	NAME OF DATABASE
1	1	Exploring retrie	1	0	0	Yes	Short-text disci	WoS
2	2	An Accurate an	1	0	1	Yes	Uses LLMs and	WoS
3	3	Retrieval-augm	1	0	0	Yes	Bibliographic Te	WoS
4	4	Exploring the a	1	0	1	Yes	Citation contex	WoS
5	5	Recent Trends	0	1	1	Yes	Bibliometric to	WoS
6	6	Supporting You	0	0	1	Yes	Focuses on kno	WoS
7	7	Investigating to	0	0	1	Yes	Evaluates topic	WoS
8	8	Topic Modelling in Hindi Using BERT, LDA, LSA and NMF Approc				No		WoS
9	9	A Systematic Li	0	0	1	Yes	Systematic revi	WoS
10	10	Exploring the R	1	1	1	Yes	Uses Large Lan	WoS
11	11	Bibliographic network enhanced local citation recommendatio				No		WoS
12	12	Ontology-base	1	0	1	Yes	Uses ontology-	WoS
13	13	Conversational	1	0	1	Yes	Discusses conv	WoS
14	14	When text mini	0	0	1	Yes	Reviews biblior	WoS

Figure 9: Screenshot of the Excel dataset used for recording keyword occurrences across three screening levels: AI-generated, manual screening till abstract, and manual screening after abstract.

6.4 GitHub Repository

All scripts, datasets, and code developed during this internship were version-controlled and made publicly available through a dedicated GitHub repository. The repository includes the R scripts used to generate the Sankey diagram, the Python scripts used for bibliometric data processing and keyword frequency analysis, and the LaTeX source files for this report. The repository is accessible at the following URL: https://github.com/AAMIRQADIR-79/Epistemic-Change-Detection-Using-LLMs . Figure 10 shows a screenshot of the GitHub repository created for this project.

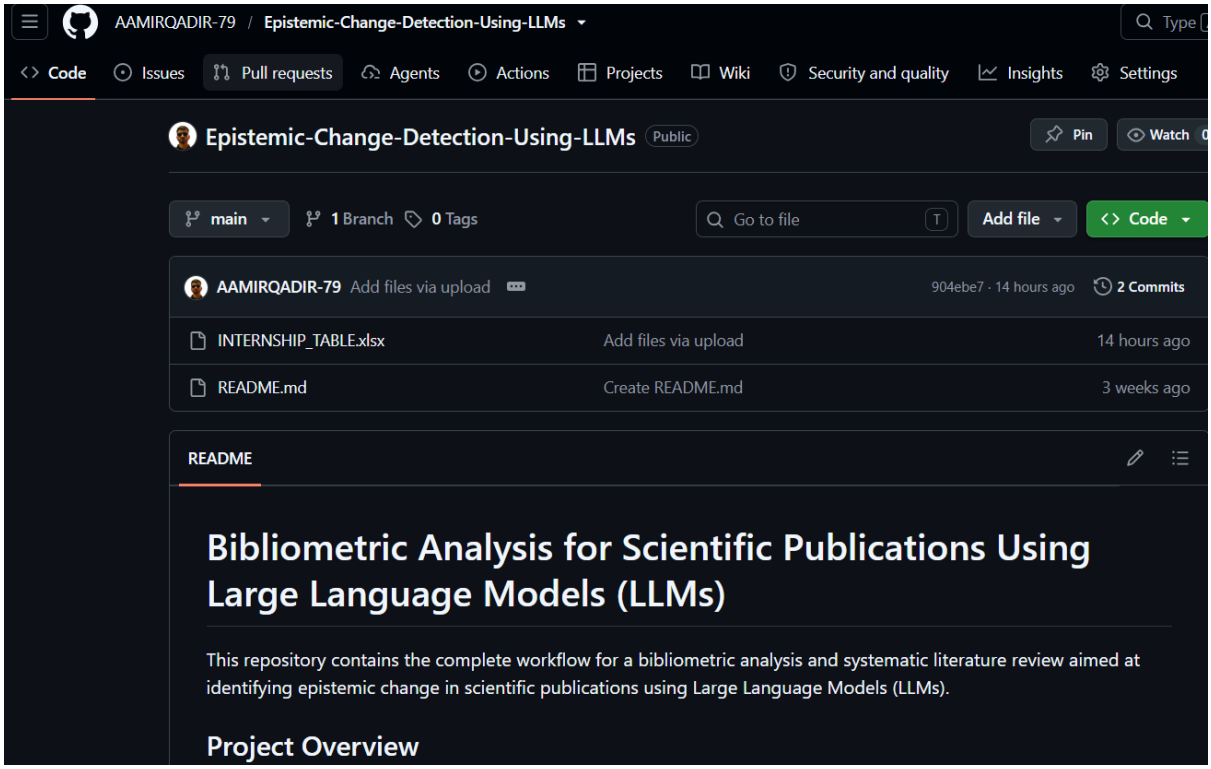


Figure 10: Screenshot of the GitHub repository `Epistemic-Change-Detection-Using-LLMs` containing all scripts, datasets, and source files developed during the internship.

References

[1] Mithila Arman, Md Farhad Kabir, Fakir Mashuque Alamgir, Kazi Aklima, Mohammad Kasedullah, Efaz Kabir, Abdullah Rakib Akand, Abdullah Al Imtihan, and Miguel Ángel Rodríguez Sánchez. Forecasting the evolution of scientific knowledge: A dynamic knowledge graph approach integrating temporal embeddings and large language models. *Machine Learning*, 115(6):148, 2026.

[2] Sören Auer, Dante AC Barone, Cassiano Bartz, Eduardo G Cortes, Mohamad Yaser Jaradeh, Oliver Karras, Manolis Koubarakis, Dmitry Mouromtsev, Dmitrii Pliukhin, Daniil Radyush, et al. The sciqa scientific question answering benchmark for scholarly knowledge. *Scientific Reports*, 13(1):7240, 2023.

[3] Iz Beltagy, Kyle Lo, and Arman Cohan. Scibert: A pretrained language model for scientific text. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 3615–3620, 2019.

[4] Arman Cohan and Nazli Goharian. Scientific document summarization via citation contextualization and scientific discourse. *International Journal on Digital Libraries*, 19(2):287–303, 2018.

[5] John Dagdelen, Alexander Dunn, Sanghoon Lee, Nicholas Walker, Andrew S Rosen, Gerbrand Ceder, Kristin A Persson, and Anubhav Jain. Structured information extraction from scientific text with large language models. *Nature communications*, 15(1):1418, 2024.

[6] Laércio Dias, Martin Gerlach, Joachim Scharloth, and Eduardo G Altmann. Using text analysis to quantify the similarity and evolution of scientific disciplines. *Royal Society open science*, 5(1):171545, 2018.

[7] Soohyung Joo, Inkyung Choi, and Namjoo Choi. Topic analysis of the research domain in knowledge organization: A latent dirichlet allocation approach. *Knowledge Organization*, 45(2), 2018.

[8] Sai Koneru. Computational analysis of scientific literature for modeling knowledge evolution, linking hypotheses to evidence, and evaluating model integrity. 2026.

[9] Kathy McKeown, Hal Daume III, Snigdha Chaturvedi, John Paparrizos, Kapil Thadani, Pablo Barrio, Or Biran, Suvarna Bothe, Michael Collins, Kenneth R Fleischmann, et al. Predicting the impact of scientific concepts using full-text features. *Journal of the Association for Information Science and Technology*, 67(11):2684–2696, 2016.

[10] Sepideh Mesbah, Kyriakos Fragkeskos, Christoph Lofi, Alessandro Bozzon, and Geert-Jan Houben. Facet embeddings for explorative analytics in digital libraries. In *International conference on theory and practice of digital libraries*, pages 86–99. Springer, 2017.

[11] Mikyung Shin and John William McKenna. Exploring the research landscape on single-case design methodology using technology through text mining and large language models. *Journal of Behavioral Education*, pages 1–31, 2026.

[12] Jinkai Tao, Yubo Wang, Xiaoyu Liu, and Menglin Yang. Continuous knowledge metabolism: Generating scientific hypotheses from evolving literature. *arXiv preprint arXiv:2604.12243*, 2026.

[13] Jianping Wang, Mingtao Lu, Wanru Wang, and Xiaoling Huang. An integrated lda and scibert model for topic analysis: A case study of network science. *Sage Open*, 16(2):21582440261426946, 2026.

[14] Min-Hsien Weng, Shaoqun Wu, and Mark Dyer. Identification and visualization of key topics in scientific publications with transformer-based language models and document clustering methods. *Applied Sciences*, 12 (21):11220, 2022.

[15] Yu Zhang, Xiusi Chen, Bowen Jin, Sheng Wang, Shuiwang Ji, Wei Wang, and Jiawei Han. A comprehensive survey of scientific large language models and their applications in scientific discovery. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8783–8817, 2024.